

SOMESH PANCHAL

Machine Learning Engineer | LLM & GenAI | Inference Optimization

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PROFESSIONAL SUMMARY

Machine Learning Engineer with approx 5 years of experience building and optimizing production ML systems, with a focus on LLM inference, GenAI applications, distributed training, and model-serving infrastructure. Experienced in taking ML models from experimentation to production with measurable improvements in latency, throughput, GPU memory, training efficiency, and cost. Hands-on expertise across PyTorch, CUDA, Hugging Face Transformers, vLLM, quantization, torch.compile, FastAPI, Docker, and Linux. Built benchmarking frameworks and agentic AI applications using LangGraph, RAG, vector databases, and self-correcting workflows. Strong software engineering foundation with experience developing reliable, observable, and production-ready ML services.

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, Bash

Machine Learning & Deep Learning: PyTorch, Hugging Face Transformers, Sentence-Transformers, scikit-learn, XGBoost, LightGBM, TensorFlow

LLM & Generative AI: LLM Fine-Tuning, LoRA, QLoRA, PEFT, Prompt Engineering, RAG, Embeddings, Semantic Search, AI Agents, Tool Use, LangGraph, LangChain

Inference Optimization: PTQ, QAT, GPTQ, AWQ, bitsandbytes, KV Cache, Flash Attention, torch.compile, TorchScript, ONNX Runtime, TensorRT, Pruning, Knowledge Distillation

Distributed Training: PyTorch DDP, FSDP, DeepSpeed ZeRO, Mixed Precision (AMP), Gradient Accumulation, Gradient Checkpointing, Pipeline Parallelism, Tensor Parallelism

Model Serving: vLLM, Triton Inference Server, TorchServe, FastAPI, REST APIs, Request Batching, Streaming, Caching, Autoscaling

Data & Storage: FAISS, pgvector, PostgreSQL, MySQL, SQL Server, Parquet, JSON, Pandas, NumPy

MLOps & Infrastructure: Docker, Docker Compose, Git, GitHub Actions, CI/CD, MLflow, Model Registry, Prometheus, Grafana, Logging, Observability

Cloud & Compute: AWS, GCP, Azure Fundamentals, NVIDIA GPUs, CUDA, Linux

Engineering & Methodologies: Agile/Scrum, SDLC, Code Reviews, Profiling & Optimization, A/B Testing, Model Evaluation

PROFESSIONAL EXPERIENCE

Qorvo, Inc | Dallas, TX

Machine Learning Engineer | Aug 2025 – Present

- Reduced per-epoch LLM training time by 35% by implementing PyTorch Distributed Data Parallel (DDP) and mixed-precision training across multi-GPU Linux environments, accelerating experimentation for research teams.
- Reduced production inference latency by 40% with no measurable quality loss by applying post-training quantization, KV-cache reuse, and torch.compile graph optimization to PyTorch models.
- Reduced GPU memory utilization by 25% by profiling training and inference workloads with PyTorch Profiler and Nsight, then applying gradient checkpointing and selective precision casting to bottleneck layers.
- Built and deployed containerized ML model-serving APIs with FastAPI, incorporating request batching, streaming, and caching and packaging services with Docker for reproducible, autoscaling-ready deployments.
- Established P95/P99 latency, throughput, GPU utilization, and cost-per-request as production KPIs and used recurring performance reviews to identify and prioritize optimization opportunities.
- Integrated production ML pipelines with SQL and Parquet data sources and FAISS vector stores, supporting retrieval-augmented and feature-driven inference workflows for downstream product teams.
- Managed MLflow experiment tracking and Git-based CI/CD workflows for model versioning and deployment, improving collaboration between research scientists and platform engineers.

DXC Technologies | India

Machine Learning Engineer | Dec 2019 – Jul 2023

- Developed machine learning pipelines using Python, Pandas, NumPy, and SQL to preprocess, validate, and transform large-scale structured datasets for predictive analytics and model development.
- Built feature-engineering workflows and automated data preparation processes, reducing manual effort and improving model training efficiency.
- Developed and optimized analytical SQL queries for extracting, profiling, and validating data from enterprise systems and relational databases.

- Assisted in training and evaluating machine learning models using scikit-learn and Python-based frameworks, improving prediction accuracy through iterative feature selection and hyperparameter tuning.
- Implemented data quality validation checks and automated monitoring processes to improve reliability and consistency of downstream ML workflows.
- Developed reusable Python modules and batch-processing utilities on Linux environments, supporting scalable data-processing and model-development activities.
- Collaborated with data engineers, software developers, and business stakeholders in Agile/Scrum teams to deliver analytics and machine learning solutions aligned with business requirements.
- Maintained code through Git-based version control, participated in peer reviews, and contributed to CI/CD-oriented development practices.

PROJECTS

LLM Inference Optimization Benchmark Suite

PyTorch | vLLM | ONNX Runtime | GPTQ | AWQ | FastAPI | Docker

<https://github.com/Somesh-6711/LLM-Inference-Optimization-Benchmark-Suite>

- Designed and developed a reproducible benchmarking framework comparing six inference configurations for Qwen2.5-1.5B-Instruct, including vanilla PyTorch, torch.compile, ONNX Runtime, GPTQ 4-bit, AWQ 4-bit, and vLLM with PagedAttention and continuous batching.
- Evaluated P50/P95/P99 latency, tokens-per-second throughput, peak GPU memory, and output quality using matched workloads and a held-out evaluation set.
- Achieved up to approximately 3x throughput improvement and 60% GPU memory reduction versus the vanilla PyTorch baseline using vLLM and 4-bit quantization, with negligible quality degradation on the evaluation set.
- Implemented GPTQ/AWQ post-training quantization, KV-cache reuse, and Flash Attention optimization paths and used PyTorch Profiler and nvidia-smi to analyze performance bottlenecks.
- Deployed the selected configuration through a FastAPI service with streaming and batching, packaged the application using Docker Compose, and implemented GitHub Actions CI/CD for linting, testing, build, and deployment.
- Created a Grafana dashboard tracking latency, throughput, and GPU utilization and documented the benchmark methodology and results for reproducibility.

Multi-Agent SQL Analyst

LangGraph | PostgreSQL | FastAPI | Streamlit | Docker

<https://github.com/Somesh-6711/Multi-Agent-SQL-Analyst>

- Built an end-to-end multi-agent LLM application that converts natural-language business questions into validated SQL, executes queries against PostgreSQL, self-corrects execution errors, and generates written reports with charts.
- Architected four cooperating LangGraph agents—Planner, Executor, Critic, and Reporter—to decompose questions, execute SQL, validate results, retry failed queries, and generate analytical summaries.
- Evaluated the workflow using a Spider benchmark subset and measured execution accuracy, retry rate, and average tokens and latency per query.
- Implemented a Critic-based self-correction workflow and evaluated its performance against a single-shot baseline.
- Integrated Groq's hosted Llama-3.3-70B endpoint with structured outputs, prompt templates, and SQL safety controls including read-only database access, query allowlists, and timeout limits.
- Built a FastAPI backend and Streamlit interface, containerized the application and PostgreSQL database with Docker Compose, and configured GitHub Actions for automated testing, linting, and deployment.

EDUCATION

Master of Science (M.S.), Computer & Information Science

Southern Arkansas University | Magnolia, AR | Aug 2023 – May 2025

Bachelor of Engineering (B.E.), Computer Engineering

Gujarat Technological University | Gujarat, India | Aug 2018 – Jul 2022